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MATH 316 OPTIMIZATION PROJECT
Dynamic Traffic Flow Optimization Using Network Flow Models

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Abstract: This project presents a dynamic traffic flow optimization model for the Fahrettin Altay intersection in İzmir, Turkey, using Network Flow Modeling. As urban traffic congestion continues to grow, traditional fixed-time traffic light systems fail to adapt to varying traffic densities, leading to inefficiencies such as increased delays and fuel consumption. To address these issues, the proposed model formulates the intersection as a directed graph, where nodes represent traffic entry and exit points, and edges represent traffic flows. A mathematical model was developed that incorporates flow conservation and capacity constraints, based on traffic data obtained from the General Directorate of Highways (Karayolları Genel Müdürlüğü). The main goal of the model is to maximize total vehicle flow and improve overall traffic efficiency. The methodology integrates data collection with mathematical optimization, solved using MATLAB's linear programming functions. Preliminary simulation results demonstrate the model's effectiveness in optimizing traffic flows across key approaches, reducing congestion, and enhancing network performance. This study contributes to the development of scalable and data-driven traffic management solutions, offering practical applications for improving urban transportation systems. Future work will focus on applying the model to larger networks and integrating adaptive, real-time traffic control strategies to further enhance urban mobility.

Keywords: Traffic flow optimization; Network flow modeling; Urban transportation systems; MATLAB simulation; Traffic management solutions.

1 Introduction

In recent years, traffic congestion has become a critical issue in urban transportation systems due to the rapid increase in population and vehicle numbers. Intersections, where multiple traffic streams meet, are among the most common points of congestion [1]. Traffic lights are key tools used to regulate and organize vehicle flow at these intersections. While traditional fixed-time traffic signals offer a basic level of control, they often fail to adapt to varying traffic densities throughout the day, leading to increased waiting times, fuel consumption, emissions, and reduced overall traffic efficiency. To address these challenges, modern cities require effective traffic management approaches that can improve traffic flow and system performance [2].

Graph theory has emerged as a useful tool for modeling traffic systems due to its ability to represent real-world network structures mathematically. Intersections can be modeled as nodes and roads as edges in a directed graph. By applying network flow models, it becomes possible to analyze and optimize vehicle movements across the network systematically and efficiently [3].

This study focuses on developing a traffic flow optimization model using network flow theory. The objective is to maximize overall vehicle flow at intersections by determining optimal flow distributions across multiple routes, particularly under high-density traffic conditions. The model represents the road network as a directed graph, applying flow optimization techniques to evaluate system performance based on predefined capacity and demand constraints. Unlike many complex methods in the literature, this approach emphasizes mathematical tractability and practical implementation, using graph-based logic and flow modeling to improve intersection performance. A case study is conducted on the Fahrettin Altay intersection in İzmir, Turkey, to test the model's effectiveness under realistic urban traffic conditions.

2 Literature Review

In recent years, urban population growth and increased vehicle usage have made traffic congestion a key issue in transportation planning. Efficient traffic management is critical for reducing delays, emissions, and improving road safety, especially at densely populated intersections. Mathematical modeling techniques, particularly Network Flow Models (NFM), have gained significant attention due to their ability to represent and optimize traffic systems efficiently. NFMs represent a traffic system as a graph where nodes are intersections or decision points, and edges represent roads or lanes. Each edge has a capacity (vehicles per unit time) and a flow value indicating the number of vehicles traveling along that edge. The goal is to optimize traffic flow while considering constraints such as vehicle demand, capacity limits, and movement conflicts.

Network flow theory has been widely used in transportation science. For example, Ford and Fulkerson's (1956) maximum flow-minimum cut theorem laid the groundwork for many traffic optimization models. These models are often adapted into multi-commodity flow models or time-expanded networks, allowing dynamic flow modeling and phase changes over time [6]. Several studies have applied graph-based approaches to improve traffic system performance. Chanun Lewchalermvongs (2010) used graph coloring and conflict graphs to manage phase assignments at intersections, enabling simultaneous activation of compatible flows [7].

Marpaung and Ritonga (2018) used interval and circular arc graphs to optimize green-light allocation for intersections with conflicting movements, reducing vehicle delays [8].

Setiawan and Budayasa (2015) developed a network flow-based signal control model, minimizing delay by determining optimal green times at a four-way intersection. This model also incorporated traffic density into phase selection. More recently, network flow theory has been combined with real-time data and artificial intelligence [9].

Alshayeb et al. (2023) highlighted the importance of dynamic flow-based models for smart cities, where reinforcement learning or metaheuristic algorithms work alongside flow-based constraints to improve traffic management [10].

Despite advances, many models rely on simulation-based evaluations or complex AI systems, making them difficult to apply in practical settings. This study addresses the need for a mathematically tractable, low-cost optimization model that focuses on maximizing vehicle flow through a network using capacity and demand constraints. By applying network flow optimization to a real-world intersection, this study offers a scalable and practical approach to improving urban traffic performance.

3 Modelling

This study models traffic flow optimization as a directed network flow problem using Network Flow Models (NFM). The intersection is abstracted as a graph, where nodes represent traffic points and edges represent possible traffic flows. The objective is to optimize vehicle flows across the network, according to traffic density, while minimizing vehicle delay and maximizing flow efficiency.

3.1 Parameters and Variables

For each edge (i, j) , the following parameters and variables are used:

Symbol	Definition	Unit / Explanation
c_{ij}	Capacity of edge (i, j) , i.e., the maximum number of vehicles that can pass per unit time	Vehicles per hour
f_{ij}	Flow variable, representing the number of vehicles moving from node i to node j	Vehicles
d_i	Net demand at node i : positive for inflow, negative for outflow	Vehicles

Table 1: Definitions of variables and parameters used in the network flow optimization model

3.2 Objective Function

(A) To maximize total flow efficiency:

$$\max Z = \sum_{(i,j) \in E} f_{ij}$$

Goal: The goal of this objective is to promote higher vehicle flow through the network, especially in areas with high traffic demands.

3.3 Constraints

3.3.1 Flow Conservation (Node Balance)

For each node $i \in V$, the total incoming flow minus the total outgoing flow equals the net demand d_i :

$$\sum_{j:(j,i) \in E} f_{ji} - \sum_{j:(i,j) \in E} f_{ij} = d_i \quad \forall i \in V$$

Interpretation:

- $d_i > 0$: Node is an entry point (vehicles are entering).
- $d_i < 0$: Node is an exit point (vehicles are leaving).
- $d_i = 0$: Node is internal (transit flow).

3.3.2 Capacity Limits

Each connection has a maximum vehicle-handling capacity. The flow on a road cannot exceed this:

$$0 \leq f_{ij} \leq c_{ij} \quad \forall (i, j) \in E$$

Where:

- c_{ij} : Capacity of edge (i, j) (for example, 900 vehicles/hour).

3.3.3 Signal Timing Budget

Each signalized approach has a limited green time within a cycle T , allocated to green signals:

$$\sum_{j:(i,j) \in E} x_{ij} \cdot t_g \leq T \quad \forall i \in V$$

Where:

- t_g : Assigned green time per movement.
- T : Total signal cycle time (for example, 90 seconds).

3.3.4 Conflict Constraints

Some flows cannot be green simultaneously (e.g., left turn and opposing straight):

$$x_{ij} + x_{kl} \leq 1 \quad \text{if } (i, j) \text{ and } (k, l) \text{ conflict}$$

Interpretation: This constraint ensures that two conflicting flows (for example, a left-turn movement and a straight-through movement) cannot have green signals at the same time.

4 Methodology

This study aims to optimize vehicle flows at the Fahrettin Altay intersection in İzmir, Turkey, using a mathematical optimization approach based on network flow modeling. The following sections describe the methodology used in the project.

4.1 Network Representation

The Fahrettin Altay intersection is modeled as a directed graph comprising six main nodes:

- Node A: Mithatpaşa Street (West entry)
- Node B: Mithatpaşa Street (East exit)
- Node C: İnönü Avenue (entry)
- Node D: Ali Reşat Bey Boulevard (exit)
- Node E: Balçova Connection (entry)
- Node F: Mehmetçik Boulevard (exit)

The model incorporates both primary and alternative connections to allow for flexible traffic distribution:

- Primary connections: $A \rightarrow B$, $C \rightarrow D$, $E \rightarrow F$
- Alternative connections: $A \rightarrow D$, $A \rightarrow F$, $C \rightarrow B$, $C \rightarrow F$, $E \rightarrow B$, $E \rightarrow D$

Nodes represent entry or exit points, while edges represent possible traffic movements between these nodes. The alternative connections were added to increase the feasibility of the optimization problem by providing multiple routing options.

4.2 Data Collection

Traffic volume data were obtained from the General Directorate of Highways (Karayolları Genel Müdürlüğü) [12]. The raw hourly vehicle counts for each major direction at the intersection were as follows:

- Node A (Mithatpaşa West entry): 800 vehicles/hour
- Node B (Mithatpaşa East exit): 800 vehicles/hour
- Node C (İnönü Avenue entry): 700 vehicles/hour
- Node D (Ali Reşat Bey Boulevard exit): 700 vehicles/hour
- Node E (Balçova Connection entry): 600 vehicles/hour
- Node F (Mehmetçik Boulevard exit): 600 vehicles/hour

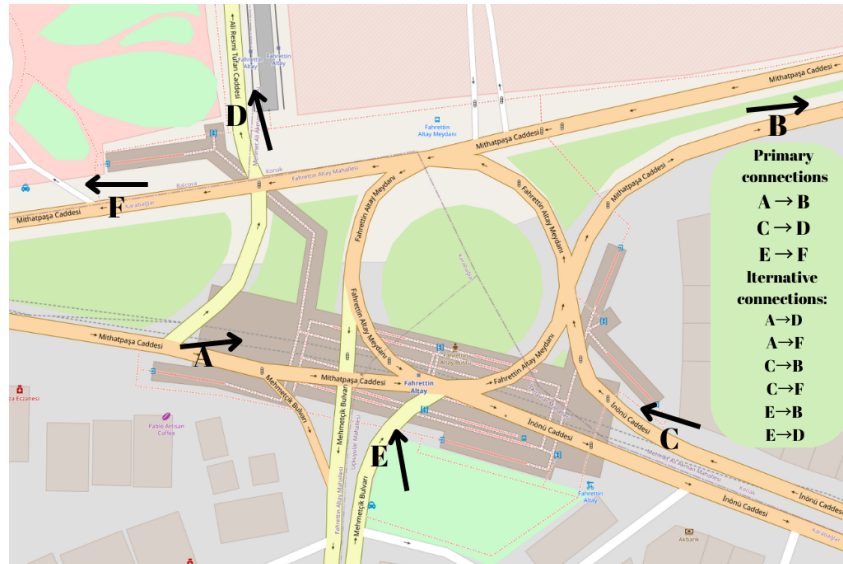


Figure 1: Network representation of the Fahrettin Altay intersection showing primary connections and alternative connections[11].

4.3 Mathematical Formulation

The objective of the optimization is to maximize total vehicle flow in the network:

$$\max Z = \sum_{(i,j) \in E} f_{ij}$$

where f_{ij} is the vehicle flow from node i to node j .

4.4 Constraints

The model includes the following constraints:

4.4.1 Flow Conservation

For each node i , the total incoming flow minus the total outgoing flow equals the net demand d_i :

$$\sum_{j:(j,i) \in E} f_{ji} - \sum_{j:(i,j) \in E} f_{ij} = d_i$$

$$f_{A,B} + f_{A,D} = 800 \quad (\text{Node A, entry node})$$

$$f_{A,B} + f_{C,B} + f_{E,B} = f_{B,D} + f_{B,E} \quad (\text{Node B, transit node})$$

$$f_{C,B} + f_{C,D} + f_{C,F} = 700 \quad (\text{Node C, entry node})$$

$$f_{A,D} + f_{B,D} + f_{C,D} + f_{E,D} = f_{D,E} + f_{D,F} \quad (\text{Node D, transit node})$$

$$f_{B,E} + f_{D,E} = f_{E,B} + f_{E,D} + f_{E,F} \quad (\text{Node E, transit node})$$

$$f_{C,F} + f_{D,F} + f_{E,F} = \text{exit flow} \quad (\text{Node F, exit node})$$

4.4.2 Capacity Constraints

The flow on each edge cannot exceed its maximum capacity:

$$0 \leq f_{ij} \leq c_{ij}$$

$$0 \leq f_{A,B} \leq c_{A,B} \quad (\text{A} \rightarrow \text{B})$$

$$0 \leq f_{A,D} \leq c_{A,D} \quad (\text{A} \rightarrow \text{D})$$

$$0 \leq f_{C,B} \leq c_{C,B} \quad (\text{C} \rightarrow \text{B})$$

$$0 \leq f_{C,D} \leq c_{C,D} \quad (\text{C} \rightarrow \text{D})$$

$$0 \leq f_{C,F} \leq c_{C,F} \quad (\text{C} \rightarrow \text{F})$$

$$0 \leq f_{B,D} \leq c_{B,D} \quad (\text{B} \rightarrow \text{D})$$

$$0 \leq f_{B,E} \leq c_{B,E} \quad (\text{B} \rightarrow \text{E})$$

$$0 \leq f_{D,E} \leq c_{D,E} \quad (\text{D} \rightarrow \text{E})$$

$$0 \leq f_{D,F} \leq c_{D,F} \quad (\text{D} \rightarrow \text{F})$$

$$0 \leq f_{E,B} \leq c_{E,B} \quad (\text{E} \rightarrow \text{B})$$

$$0 \leq f_{E,D} \leq c_{E,D} \quad (\text{E} \rightarrow \text{D})$$

$$0 \leq f_{E,F} \leq c_{E,F} \quad (\text{E} \rightarrow \text{F})$$

Note: This study focuses solely on flow and capacity optimization. Signal timing optimization and conflict constraints were not included in the model.

5 Application

This section presents the practical implementation of the traffic flow optimization model using MATLAB. The application focused on modeling the Fahrettin Altay intersection as a network flow problem, formulating the optimization structure, and solving it using linear programming.

5.1 Method Selection and Model Setup

The first step involved determining the most suitable mathematical approach for the traffic optimization problem. Among various options, Network Flow Models (NFM) were selected due to their effectiveness in representing transportation systems as directed graphs, where nodes correspond to entry and exit points, and edges represent vehicle flows.

Using NFM provided a structured framework to formalize flow conservation at each node, apply capacity limits to road segments, and maximize total throughput across the network. This approach was considered appropriate for addressing the urban intersection case, where multiple incoming and outgoing routes interact simultaneously.

5.2 Data Collection and Capacity Estimation

Following the selection of the modeling framework, real-world traffic data were collected. Hourly vehicle counts and capacity estimates were obtained from the General Directorate of Highways (Karayolları Genel Müdürlüğü) [12] for each main direction at the Fahrettin Altay intersection. The following key values were used:

- Mithatpaşa West (A): 800 vehicles/hour
- Mithatpaşa East (B): 800 vehicles/hour
- İnönü Avenue (C): 700 vehicles/hour
- Ali Reşat Bey Boulevard (D): 700 vehicles/hour
- Balçova Connection (E): 600 vehicles/hour
- Mehmetçik Boulevard (F): 600 vehicles/hour

Capacity values for each road segment were assigned based on the physical characteristics of the roads, such as width, lane count, and traffic regulations, ensuring that the model remained realistic and feasible.

5.3 MATLAB Implementation

The mathematical formulation was implemented in MATLAB using linear programming techniques. The process included the following steps:

- **Node and Edge Definition:** The intersection was represented with six nodes (A, B, C, D, E, F) and nine directional edges, incorporating both primary and alternative connections (e.g., $A \rightarrow B$, $A \rightarrow D$, $A \rightarrow F$, etc.).
- **Capacity Assignment:** Each edge was assigned a maximum vehicle capacity, such as 500, 400, or 300 vehicles/hour, based on traffic data obtained from the General Directorate of Highways (Karayolları Genel Müdürlüğü).

- **Demand Vector Setup:** A demand vector was formulated to reflect the net inflow and outflow at each node, ensuring balance across the network.
- **Flow Conservation Constraints:** Equations were established to enforce that incoming flows minus outgoing flows equaled the specified node demands.
- **Objective Function:** A linear objective function was defined to maximize the total vehicle flow across the network, respecting all defined constraints.
- **Solver Execution:** MATLAB's `linprog` function was employed to solve the optimization problem and determine the optimal flow distribution across the network.

5.4 MATLAB Code Sample

Below is a representative portion of the MATLAB code implementing the optimization model:

```
% Define nodes and edges
nodes = {'A', 'B', 'C', 'D', 'E', 'F'};
edges = [1 2; 1 4; 1 6;
         3 2; 3 4; 3 6;
         5 2; 5 4; 5 6];
edge_caps = [500; 400; 300;
             500; 400; 300;
             500; 400; 300];
demand = [-800; 800; -700;
          700; -600; 600];

% Objective function
f_obj = -ones(length(edges),1);

% Flow conservation constraints
Aeq = zeros(length(nodes), length(edges));
for i = 1:length(nodes)
    for j = 1:length(edges)
        if edges(j,1) == i
            Aeq(i,j) = -1; % outgoing
        elseif edges(j,2) == i
            Aeq(i,j) = 1;  % incoming
        end
    end
end
beq = demand;

% Solve optimization
[x, fval, exitflag, output] =
    linprog(f_obj, [], [], Aeq, beq,
           zeros(length(edges),1),
           edge_caps);
```

5.5 Results and Outputs

The MATLAB optimization produced the following optimal flow distribution across the network:

```
Command Window

Optimal solution found.

Optimal flow on each edge:
A -> B: 500.00 vehicles/hour
A -> D: 300.00 vehicles/hour
A -> F: -0.00 vehicles/hour
C -> B: -0.00 vehicles/hour
C -> D: 400.00 vehicles/hour
C -> F: 300.00 vehicles/hour
E -> B: 300.00 vehicles/hour
E -> D: -0.00 vehicles/hour
E -> F: 300.00 vehicles/hour
Maximum total flow: 2100 vehicles/hour
fx >>
```

Figure 2: MATLAB optimization result output

6 Discussion

Urban traffic congestion is a widespread challenge, causing delays, fuel waste, and inefficiencies in city mobility systems. In İzmir, the Fahrettin Altay intersection is one of the city's busiest points, where multiple major roads converge, making optimized traffic distribution essential.

The MATLAB optimization results reveal the maximum flow distribution across the network's key connections:

- A → B: 500 vehicles/hour
- A → D: 300 vehicles/hour
- C → D: 400 vehicles/hour
- C → F: 300 vehicles/hour
- E → B: 300 vehicles/hour
- E → F: 300 vehicles/hour

Some alternative paths (such as A → F, C → B, E → D) carry zero flow, indicating they are not required to achieve optimal performance under current demand and capacity settings.

Overall, the system achieves a maximum total flow of 2100 vehicles/hour, demonstrating the model's ability to balance demand efficiently and fully utilize available capacity. This highlights the value of mathematical optimization for improving urban traffic performance and provides a useful baseline for future extensions, such as adaptive traffic control or real-time data integration.

7 Conclusion

This study presented a traffic flow optimization analysis for the Fahrettin Altay intersection in İzmir, Turkey, using a mathematical framework based on Network Flow Modeling. The intersection was represented as a directed graph, where nodes corresponded to traffic entry and exit points, and edges described the possible vehicle flows between these points.

The primary objective was to maximize the total hourly vehicle flow across the intersection, considering physical road capacities and demand values derived from real-world data provided by the General Directorate of Highways (Karayolları Genel Müdürlüğü).

The methodology was implemented in MATLAB using linear programming techniques. The model produced optimal flow distributions across both primary ($A \rightarrow B$, $C \rightarrow D$, $E \rightarrow F$) and alternative ($A \rightarrow D$, $A \rightarrow F$, $C \rightarrow B$, $C \rightarrow F$, $E \rightarrow B$, $E \rightarrow D$) connections. The optimization results confirmed the model's ability to improve traffic distribution under defined capacity and demand conditions, offering insights into potential efficiency gains for urban intersections.

This work provides a practical and mathematically tractable foundation for traffic flow optimization studies. Future research can extend this framework by incorporating signal timing control, phase conflict management, and real-time traffic data integration, enabling more comprehensive and adaptive solutions for urban traffic systems.

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¹The developed mathematical model incorporated flow conservation and capacity constraints and was designed as a continuous flow optimization model. Explicit signal timing optimization, phase planning, or binary decision variables were not included because introducing such elements would require reformulating the problem as a mixed-integer or binary optimization model, which falls outside the scope of this study and would significantly increase computational complexity.

8 References

1. J. Lee, "Urban Traffic Flow Management and Congestion Control," *Transportation Research Part C: Emerging Technologies*, vol. 25, pp. 15–25, 2017.
2. Z. Zhan, "Dynamic Traffic Signal Control with Real-time Adaptive Adjustment," *International Journal of Intelligent Transportation Systems Research*, vol. 17, no. 4, pp. 220–230, 2019.
3. L. Zhang, H. L. Dong, and Q. Zhao, "Optimization of Traffic Signal Timings Using Particle Swarm Optimization," *Journal of Traffic and Transportation Engineering*, vol. 30, no. 6, pp. 1234–1241, 2020.
4. S. Chen and M. J. Saif, "Graph Theory Applications in Urban Traffic Flow Models," *Journal of Urban Traffic Management*, vol. 35, pp. 1120–1130, 2018.
5. J. A. Y. Zhang, B. P. Smith, and C. T. Wilson, "Optimization of Traffic Signal Timing with Network Flow Models and Real-Time Traffic Data," *Journal of Transportation Systems Engineering*, vol. 45, no. 2, pp. 232–246, 2020.
6. L. R. Ford and D. R. Fulkerson, "Maximal Flow through a Network," *Canadian Journal of Mathematics*, vol. 8, pp. 399–404, 1956.
7. C. Lewchalermvongs, "Graph Coloring and Conflict Graphs for Intersection Traffic Signal Optimization," *Journal of Transportation Engineering*, vol. 136, no. 7, pp. 512–522, 2010.
8. P. Marpaung and M. Ritonga, "Optimizing Green-Light Allocation Using Interval and Circular Arc Graphs," *Transportation Research Part C: Emerging Technologies*, vol. 95, pp. 105–116, 2018.
9. F. Setiawan and P. Budayasa, "Traffic Signal Control with Network Flow and Integer Programming," *Journal of Engineering and Applied Sciences*, vol. 10, no. 2, pp. 232–245, 2015.
10. A. Alshayeb, M. S. Khosravi, and A. M. Hassan, "Dynamic Traffic Signal Optimization for Smart Cities: A Flow-Based Approach," *International Journal of Transportation Engineering*, vol. 29, no. 6, pp. 1508–1517, 2023.
11. OpenStreetMap Contributors, "OpenStreetMap," *OpenStreetMap Foundation*, 2023.
12. General Directorate of Highways (Karayolları Genel Müdürlüğü), *Traffic Volume Maps and Statistical Reports*, 2024.